Utjecaj vrsta povratnih veza na svojstva pojačala

Za jake povratne veze $\rightarrow \beta A \gg 1$

Naponska - serijska

$$A_{vf} = \frac{A_v}{1+\beta A_v} \rightarrow \frac{1}{\beta}$$

$$R_{ulf} = R_{ul}(1+\beta A_v) \rightarrow \infty$$

$$R_{izf} = \frac{R_{iz}}{1+\beta A_v} \rightarrow 0$$

Strujna - serijska

$$G_{Mf} = \frac{G_M}{1+\beta G_M} \rightarrow \frac{1}{\beta}$$

$$R_{ulf} = R_{ul}(1+\beta G_M) \rightarrow \infty$$

$$R_{izf} = R_{iz}(1+\beta G_m) \rightarrow \infty$$

Strujna - paralelna

$$A_{If} = \frac{A_I}{1+\beta A_I} \rightarrow \frac{1}{\beta}$$

$$R_{ulf} = \frac{R_{ul}}{1+\beta A_I} \rightarrow 0$$

$$R_{izf} = R_{iz}(1+\beta A_i) \rightarrow \infty$$

Naponska - paralelna

$$R_{Mf} = \frac{R_M}{1+\beta R_M} \rightarrow \frac{1}{\beta}$$

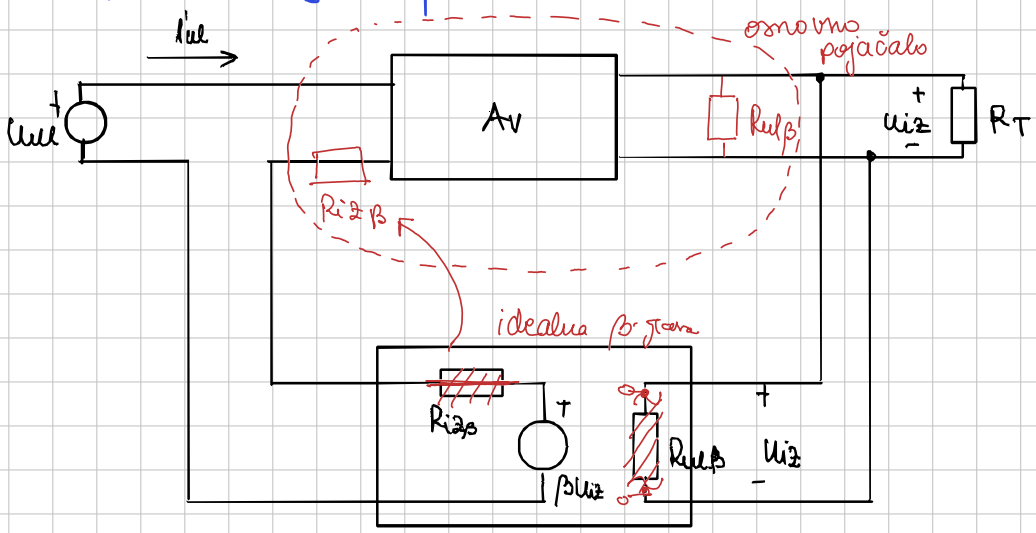
$$R_{ulf} = \frac{R_{ul}}{1+\beta R_M} \rightarrow 0$$

$$R_{izf} = \frac{R_{iz}}{1+\beta R_m} \rightarrow 0$$

Ypdf postupka analize PV $\rightarrow$ dio "alabahtera".

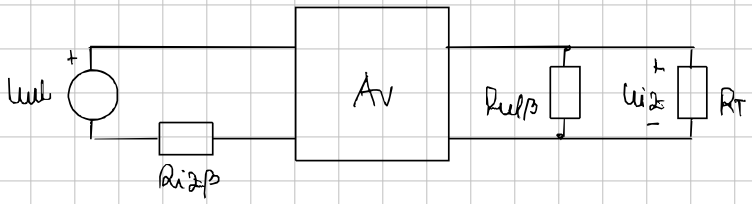
Određivanje A-grane

• naponska - serijska povratna veza

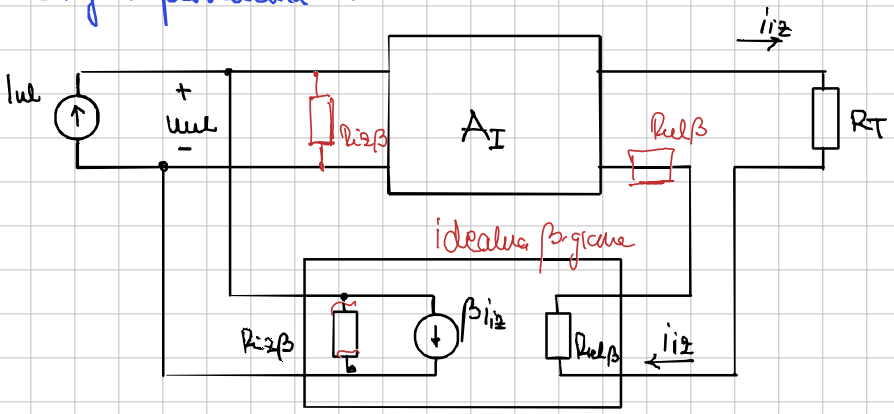


ulazni krug: uz $u_{iz}=0 \rightarrow \beta u_{iz}=0 \rightarrow R_{iz\beta}$ u ulaznom krugu

izlazni krug: uz $i_{ul}=0 \rightarrow$ prekid ulaznog kruga $\rightarrow R_{ul\beta}$ u izlaznom krugu

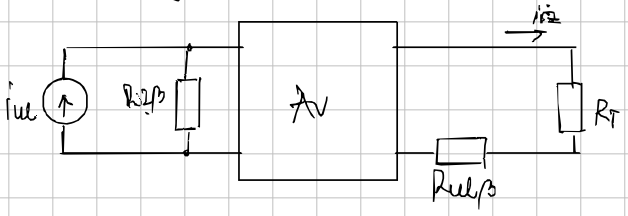


• strujna paralelna PV

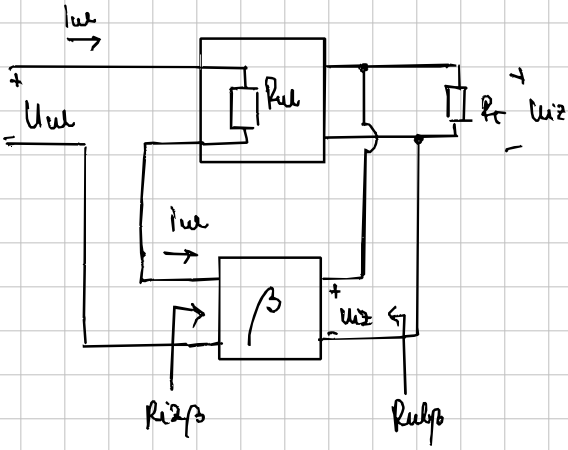


ul. krug: uz $i_z=0 \rightarrow \beta i_{iz}=0 \rightarrow R_{iz\beta}$ u ulaznom krugu

Izlazni krug: $u_z=0 \rightarrow$ k.s. ul. kruga $\rightarrow R_{ul\beta}$ u izlaznom krugu



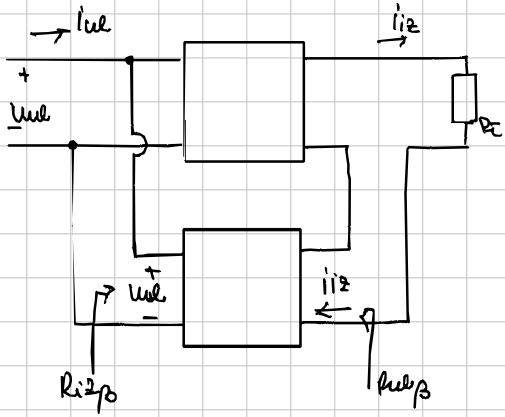
• naponska-serijska:



izlazni krug: odredivanje UNIET: iul = ∅

ulazni krug: odredivanje UNIET: uiz = ∅

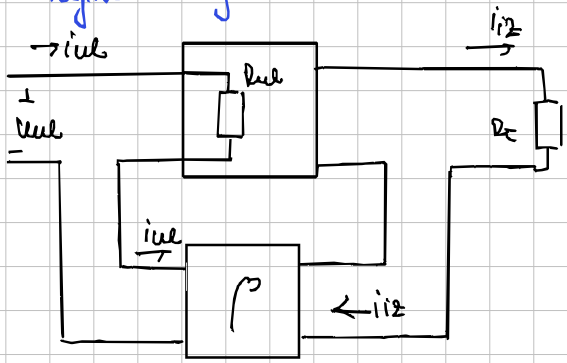
• strujna - paralelna



izlazni: uul = ∅

ulazni: iiz = ∅

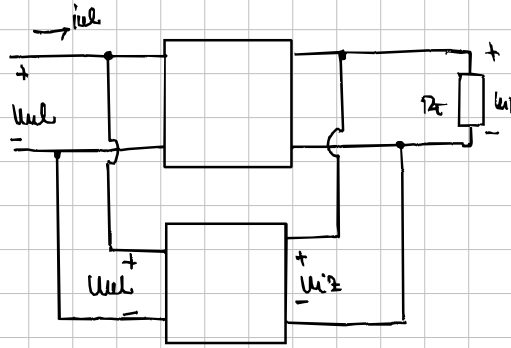
• strujno - serijska:



izlazni krug: iul = ∅

ulazni krug: iiz = ∅

• naponska - paralelna



izlazni: uul = ∅

ulazni: uiz = ∅